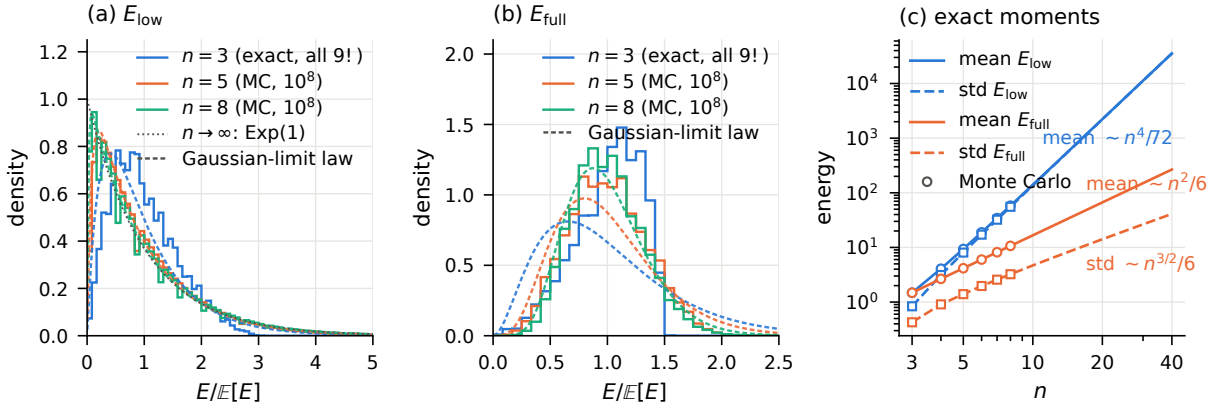
(a) E_{low}
density
$E/\mathbb{E}[E]$
$n = 3$ (exact, all 9!)
$n = 5$ (MC, 10^8)
$n = 8$ (MC, 10^8)
$n \to \infty$: Exp(1)
Gaussian-limit law
(b) E_{full}
density
$E/\mathbb{E}[E]$
$n = 3$ (exact, all 9!)
$n = 5$ (MC, 10^8)
$n = 8$ (MC, 10^8)
Gaussian-limit law
(c) exact moments
energy
n
mean E_{low}
std E_{low}
mean E_{full}
std E_{full}
Monte Carlo
mean $\sim n^4/72$
mean $\sim n^2/6$
std $\sim n^{3/2}/6$